NATIONAL UNIVERSITY OF SINGAPORE

Department of Statistics and Data Science

DSA1101 Introduction to Data Science

(Semester 2 AY2025/2026)

Midterm Test

Time Allowed: 80 minutes

INSTRUCTIONS TO STUDENTS

1. Students are required to complete this test individually.
2. This is an OPEN BOOK BLOCKED INTERNET test. You may refer to your lecture notes, tutorials, textbooks or any notes that you have made, but you are not allowed to perform any online search.
3. The use of offline Large Language Models (LLMs) is prohibited for this test.
4. All calculators in the List of Approved Calculators are allowed.
5. Students are required to answer ALL questions. Total mark is 60.
6. At the end of the test,
   - Copy and paste your R code into the Examplify text box. Indentation and alignment may not be retained when pasting, but that is acceptable.
   - Save an exact copy of your R file on your laptop, and upload it to Canvas/DSA1101/Assignments/Midterm Submission immediately after the test.
7. Do not modify the code in your submission file. Any difference found (except for indentation or alignment) between Examplify and Canvas submissions will be penalized.
8. **Your submission should have only one file (.R) which includes the R code and your interpretation/analysis as comments**. Make sure that there is no error when the graders open and run your code file.
9. Be sure to lay out systematically the various parts and steps in your code file by the comments in R code file.

1. (*15 points*) Data Manipulation and Exploration

The data file `volkswg.csv` contains data from a random sample of 15157 used Volkswagen car listings in the UK. The used Volkswagen cars being sold belong to cars registered from 2000 to 2020.
The given file contains columns with their names listed below.

Column's Name	Description
model	model of the car
year	year the car was registered
price	price of the car listing (in £)
transmission	transmission type (Automatic/Semi-Auto/Manual)
fuelType	type of fuel used (Petrol/Diesel/Hybrid/Other)

- please run function `setwd()` in a separate line (if you need it) when importing the data set into R.
- **the names given in bold below MUST be used in your R code**.

Load the file `volkswg.csv` into R and name it as **data**.

(a) (2 points) Write code to change all listings with **fuelType** equals to "Other" into "Hybrid" instead.

After that, report the number of cars of each fuel type.

(b) (2 points) Write code to discard all the cars that belongs to rare model (**model** that has less than 10 appearances in the whole data set).

After the removal, report the number of rows in **data**.

(c) (2 points) Consider some years: 2000, 2004, 2008, 2012, 2016 and 2020. Write code to create box plots of car's price for each of those years. Give your comments.

(d) (3 points) Write code to create a new column for **data**, named **priceHL**, which equals to "high" if the **price** of the car is above or equals to the median price, and equals to "low" otherwise.

Report the number of cars that were sold with low price.

(e) (3 points) Create a contingency table (named **tab1**) for **transmission** and **priceHL**, but only for "Automatic" and "Manual" cars.

Report the sum of 4 cells in the table **tab1**.

(f) (3 points) Using **tab1**, write one command in R to find the two probabilities below.

(i) the probability of having automatic transmission in the group of high price cars

(ii) the probability of having automatic transmission in the group of low price cars

Report the difference of the two probabilities above and interpret the meaning of that difference.

2. (*35 points*) Linear Regression and KNN

In R, run three lines of code below.

```
library(MASS)
data(birthwt)
data2 = birthwt
```

The names given in bold below MUST be used in your R code.

data2 is a data set of birth weight that studies the risk factors that may associate with low infant birth weight. **data2** will be used for this problem where the description of some variables of our concern is given below.

Column's Name	Description
bwt	birth weight (in grams)
low	1 = low birth weight (<2500 gr) and 0 = normal (≥2500 gr)
age	mother's age (in years)
lwt	mother's weight at last menstrual period (in pounds)
race	1 = white, 2 = black, 3 = other
smoke	smoking status during pregnancy, 1 = yes; 0 = no
ui	presence of uterine irritability, 1 = yes, 0 = no

Part I: Linear Model (19 points)

(**a**) (2 points) Report the number of infants with birth weight less than 2500 grams.

(**b**) (4 points) Create a box plot for the sample of birth weights. Does it show any outliers? If yes, then

(i) report the number of outliers,

(ii) report the value of those outliers,

(iii) report the information of those outliers on the race, the mother's age, and her smoking status.

(**c**) (4 points) Fit a linear regression model for the response variable **bwt**, named as **M1**, which uses input features: **age**, **lwt**, **race**, **smoke** and **ui**.

Report p-values of the regressors that are NOT significant in the model, at significance level 0.05.

(**d**) (3 points) Using **M1**, write code to predict the birth weight of two infants based on the information given below.

A: age = 21; lwt = 108, race = 1, smoke = 1, ui = 1.

B: age = 30; lwt = 107, race = 3, smoke = 0, ui = 1.

Report the predicted birth weights.

(**e**) (6 points) Using model M1, write code to predict the birth weight for all data points.

If an infant's predicted birth weight is less than 2500 grams, classify as low birth weight; otherwise, classify as normal.

Compute and report the Type 1 error rate and accuracy of model M1.

Part II: KNN (16 points)

For the questions in this Part II, **smoke** and **ui** are considered as quantitative variables.

We would want to form KNN classifiers using inputs **age**, **lwt**, **smoke** and **ui**, which help to predict whether an infant will be born with low birth weight.

Hence, **low** is the response variable with low = 1 (low birth weight) be considered as positive.

(**f**) (4 points) Write code to create a new data frame, called **data2.KNN** for which its first column is the column of the response **low**; and other columns are **age**, **lwt**, **smoke** and **ui** after standardization for all observations.

Copy and paste the first 5 rows of **data2.KNN** as comments in the answer file.

(**g**) (6 points) Run the command **set.seed(503)**.

Use all 189 observations as the train set with four standardized inputs to form the KNN classifiers where each classifier has the value of nearest neighbours k that is an odd numbers in the range from 3 to 15.

The values of Type 1 error rate and accuracy for each classifier are kept in two vectors, named **fpr** and **accuracy**, respectively.

Copy and paste the vectors **fpr** and **accuracy** into your code file.

(**h**) (2 points) Among all the values of k from 3 to 15, which value of k gives the KNN model lowest Type 1 error rate? Which value of k would you propose to use?

KNN model in comparison with linear model in Part I, which model performs better? Explain.

(**i**) (4 points) Use KNN model above with $k = 5$ to predict whether two infants A and B in Question (**d**) are born with low birth weight.

Report the answers in your code file.

Hint: You may standardize the values for these new observations using the mean and the standard deviation of all observations in the given data set.

3. (*10 points*) You are asked to help someone tracking the growth of their savings in a bank account. The person deposits the same fixed amount every month to his bank account, and earns a fixed monthly interest rate (on the updated balance). He wants to know how his savings grow over time.
Assume:
deposit = amount deposited at the end of each month.
rate = monthly interest rate (e.g. 0.12/12 = 0.01 for the rate of 1% per month)
months = number of months the person deposits money to bank account

(**a**) (5 points) Define a function, named **final_balance**, which

- takes arguments **rate**,

- assumes the initial balance is 0,

- uses a for loop,

- returns the final balance after **months**.

Note: you may define function **final_balance** with a few arguments but it at least must have **rate** as one of the arguments.

Use function **final_balance** to calculate the final balance for each of three persons below.

C: months = 36, deposit = 2000, rate = 0.06/12.

D: months = 36, deposit = 2000, rate = 0.04/12.

E: months = 36, deposit = 2000, rate = 0.03/12.

Report the results in the answer file.

(**b**) (5 points) Now the person wants to know: "How many months will it take me to reach a target balance?"

Write a function **months_to_target** that:

- takes arguments **deposit**, **rate**, and **target**,

- starts with **balance** = 0 and **months** = 0,

- uses a while loop to keep updating the balance until the balance is at least equal to **target**,

- returns the number of months it took to reach or exceed the target.

With fixed interest rate of 0.005 monthly, use function **months_to_target** to calculate the number of months for 3 cases below.

C: deposit = 2000, target = 100000

D: deposit = 2000, target = 150000

E: deposit = 3500, target = 120000

Report the results in the answer file.

END OF QUESTIONS